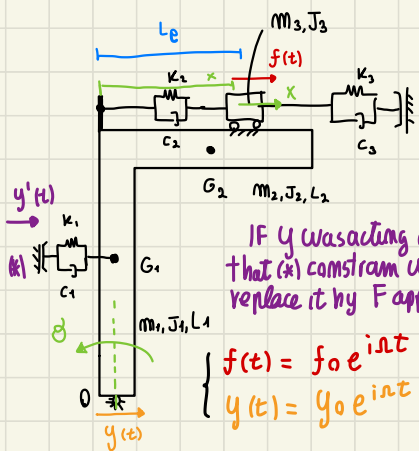


PREPARATION PROBLEM 1 EXAM FROM ESE 10

MATLAB SOL. OF ESE 10



IF y was acting on that (x) constrain we could replace it by F applied horizontally @ G1 as $(k_1 y + c_1 \dot{y}) = F = (k_1 + i \Omega c_1) y$

$[\lambda m] =$

θ	$\dot{x}$	$\dot{y}$
$-L_1/2$	0	1
0	1	0
1	0	0
$-L_1$	0	1
$L_2/2$	0	0
1	0	0
$-L_1$	1	1
$-L_e$	0	0
1	0	0

$V_{G1x}, V_{G1y}, \omega_1, V_{G2x}, V_{G2y}, \omega_2, V_{G3x}, V_{G3y}, \omega_3$

$f(t) = f_0 e^{i \Omega t}$
 $y(t) = y_0 e^{i \Omega t}$

$[\lambda F] =$

$\delta \theta$	δx	δy
$-L_1$	1	1
0	0	1
0	0	1

$\delta x_{G1}, \delta y$

$[\lambda K] =$

θ	x	y
$-L_1/2$	0	1
0	1	0
L_1	-1	-1

$\Delta p_1, \Delta p_2, \Delta p_3$

$[K_{G1}] = m_1 g$

$-L_1/2$	0	0
0	0	0
0	0	0

$[K_{G2}] = m_2 g$

$-L_1$	0	0
0	0	0
0	0	0

$[K_{G3}] = m_3 g$

$-L_1$	1	0
1	0	0
0	0	0

$[K_{del,3}] = K_3 \Delta l_{G3}$

L_0	0	0
0	0	0
0	0	0

all system matrix found by hand dummy ESE 10 (21/10) using BASIC SCRIPT

and compute all the values on the matrices

because y(t) act directly on point of the body (hinge) Vertical length we can deal with it only by including y as additional DOF

$\underline{x} = \begin{pmatrix} \theta \\ x \\ y \end{pmatrix}$ ← {constrain displacement acting on (0) HINGE}

and build all MATRICES M, C, K with 3 columns putting in it! dividing independent and constrain displacement.

QUESTION 1 EXAM

"WRITING the equation" emphasized
defining the matrix expression

after computing

$$[M] = [\lambda m]^T [m \phi h] [\lambda m] = \begin{bmatrix} \boxed{} & \boxed{} \\ \boxed{} & \boxed{} \end{bmatrix} \begin{matrix} M_{FF} (2 \times 2) & M_{FC} (2 \times 1) \\ M_{CF} (1 \times 2) & M_{CC} (1 \times 1) \end{matrix} \left. \begin{matrix} (1) \\ (2) \end{matrix} \right\}$$

and we do it also for $[C]$, $[K]$

using later this partition

to rewrite the system equation to separate the syst actual cohort (x, σ) from $y(t) \rightarrow$ resulting

actg on σ, x forces

(

$$1) M_{FF} \ddot{x}_F + C_{FF} \dot{x}_F + K_{FF} x_F = f_F - (M_{FC} \ddot{x}_C + C_{FC} \dot{x}_C + K_{FC} x_C)$$

2) NOT needed NOW on the EXERCISE ...

to find from M, C, K extract on matlab

1

to compute the constraint force horizontal actg in 0 corresponding to (H_0)

$$M_{FF} (1:2, 1:2) \uparrow \text{2 indep cohort}$$

$$M_{FF} (1:3, 1:3) \text{ if 3 indep cohort!}$$

the partition depends on how you define the x vector of cohort

• QUESTION 2 ω, X of system

! constraint displacement $y(t)$ is NOT an indep cohort for sys! $\Rightarrow$ it is an excitation

SYSTEM made out of 2 DOF so from system, after partitioning we want to find $x_F \in \mathbb{R}^2$

so find ω_1, ω_2

x_1, x_2

$$\left\{ \begin{matrix} 2 \text{ NAT FREQ} \\ 2 \text{ MODAL SHAPE} \end{matrix} \right\}$$

$$\omega \in \mathbb{R}^{\text{DOF}}$$

considering time dep. excitation removed (free system)

regardless to the number of constrain displacement as response of the system

the $y(t)$ is relevant to find I/O

where $y(t)$ is an INPUT, special time dependent excitation

$$[M_{FF}] \ddot{x}_F + [K_{FF}] x_F + [C_{FF}] \dot{x}_F = 0 \quad \text{all forcing to } Q$$

derive ω, X removing all terms of exogenous

by eig val / vect of $Eig([M_{FF}]^{-1}[K_{FF}])$

NAT FREQ:

modal shapes:

large rotation of L

$\Rightarrow$ we get $f_1 = 1.6506 \text{ Hz}$

$X_1 = (-0.9834 \quad -0.1813)^T$ small rel motion of x if slider
in phase movement
left rotation means left x motion

$f_2 = 4.3358 \text{ Hz}$

$X_2 = (-0.3324 \quad -0.9431)^T$
rotation are in phase (both same sign!)
relative movement

Rotation vs Displacement.

IT IS NOT so correct to compare the quantities...

$\Leftarrow$

doesn't affect depends on $(-)$ or $(+)$
not same sign more sliding motion of m_3 over m_2
with smaller rotation

QUESTION 3

compute FRF for $f \in (0, 0.01, 5) \text{ Hz}$ applying $f(t)$ (horiz @ m_3)
with OUTPUT θ and F_3

iterate on a for cycle the solution of complex value problem
getting output FRF points

$$\underline{x}_F = \underbrace{(-\Omega^2 M_{FF} + i\Omega C_{FF} + K_{FF})^{-1}}_{[A]} \cdot \underbrace{F_f}_{F_f}$$

iterating for various Ω 's values
saving it on an array

F : (magnitude and phase) of excitation acty on the slider m_3 horizontally, acting on (y, x)

JUST extract relevant element OUTPUT

to find F_3 we should use

Δe_3 row on $[\lambda_k]$ to define Δe_3 and find so

$$F_3 = (K_3 + i\Omega C_3) \Delta e_3 \text{ as T.F between } \Delta e_3$$

When harmonic force acty...

$\leftarrow$ and F_3 generated by SPRING + DAMPER

(ISSUE) → calculate doing this ⇒ $\underline{f}_F \approx F$ on CODE

What should we take as $\underline{F}_f$

from $\underline{f} = [\lambda_F]^T \underline{f}_{ph} = \begin{bmatrix} -L_1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}^T \begin{bmatrix} f_0 e^{i\omega t} \\ H_0 \end{bmatrix}$

defined from $(x \text{ } \theta \text{ } y)^T$

KNOWN terms, FRF by fixing $f_0 = 1$ & vary

slender force

reaction force

UNKNOWN Terms!

$$= \begin{bmatrix} -L_1 & 0 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} f_0 e^{i\omega t} \\ H_0 \end{bmatrix} = \begin{bmatrix} -L_1 \\ 1 \\ 1 \end{bmatrix} f_0 e^{i\omega t} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} H_0$$

$\underline{f}_c$

OM FRF

$e^{i\omega t}$ simplified on the equations on left / right side appearing! HARMONIC

$\underline{f}_F$ (2 Row)

PARTITION into $\underline{f}_F$ (and $\underline{f}_c$)

↑ of the 2nd syst equation related to constraint

to define

$\underline{f}_{el3}$, from $[\lambda_k]$

$\Delta l_3 = L_1 \theta - x - y$

we don't care of $y(z)$ for NOW, don't care on the displacement θ at this point

IF instead we had input as $y(z) = 1 e^{i\omega t}$

we should take also $L_1 \theta - x - 1 = \Delta l_3$ affects elongation

by unitary displacement the def. of the spring (amplitude)

at the end we get BODE

plot with an expected resonance pick @ f_1, f_2

as we cancel on $\underline{f}_{el3}$ FRF @ $f = f_2$ we have an HIGH RES PICK ≈ 30 times

• QUESTION 4 $\Delta \theta_1$ done by hand on ES10 (21/10)

with STATIONARITY OR D'ALAMBERT

↳ done on paper analytically

⇒ WRITE on MATLAB script only the final

$\Delta \theta_3$ expression that provides the solution on the code
(write also on PAPER)

• QUESTION 5 time history of V_0 (vertical constr @ 0) given

$F_{01} e^{i(\Omega_1 t + \varphi_1)}$ applied to m_3 and $y_{02} e^{i(\Omega_2 t + \varphi_2)}$ CONSTRAIN DISPLACEMENT.

↓
both inputs excitation

actg with (F_{01}, y_{02}) amplitude $\neq 1$
and phase φ_1, φ_2

applied to the hinge

⇒ OUTPUT as real
number! considering
TWO INPUTS by

applied on linearized system
equation → find time history

← USING SUPERIMPOSITION
PRINCIPLE

as SUM of the two

effects of F_0, y_0 alone applied

enumerating F_{01}, φ_1 → is NOT as issue, we JUST need
 y_{02}, φ_2 to multiply the response by
magnitude and add phase shift

do this NOT more on a for Ω cycle but

fixing $\begin{cases} \Omega_1 t + \varphi_1 \\ \Omega_2 t + \varphi_2 \end{cases}$ fixed by data

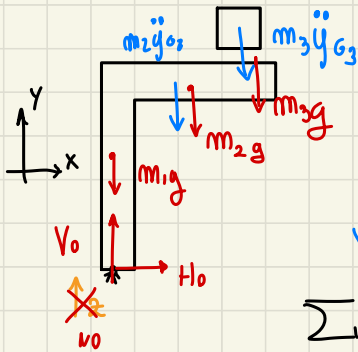
evaluate for Input $f(t)$ → so evaluate for $\underline{f} = [-L_1 \quad 1]^T F_{01} e^{i(\varphi_1 + \Omega_1 t)}$

than $y(t)$ → evaluate when $\underline{f}_c = y_{02} e^{i\varphi_2}$

↓
MATLAB

how to find V_0 vertical constraint on HINGE $\odot$

Equilibrium equation



$\left\{ \begin{array}{l} H_0 \text{ can be found by (2) equation} \\ \text{for constraints} \end{array} \right\}$

V_0 cannot be found from the syst eq.

BUT by **DALAMBERT** principle equations

(you could add a new partitioning $z=0$ and found V_0) $\rightarrow$ NOT so good!

$$\sum F_y = 0 \quad \text{from this simple equation we directly found } V_0$$

(syst)

advantage... in this way you neglect internal transmitted constraint between L BODY and SLIDER!

and taking F_y , K_1, K_2, K_3 apply forces ONLY on HORIZONTAL direction and K_2 is INTERNAL so negligible!

so we have

JUST INERTIAL forces effect and from λ in looking I ORD

TERMS as we want $\rightarrow$ looking VERTICAL speed

G_1 has NO VERT mov. $V_{G1y} = 0 \rightarrow a_{G1y} = 0$ only horiz accelerat.

While $V_{G2y} \neq 0$

$$V_0 = \underbrace{m_1 \ddot{y}_{G2} + m_3 \ddot{y}_{G3}}_{\text{dynamically fluctuating part}} + \underbrace{m_1 g + m_2 g + m_3 g}_{\text{STATIC PART, can be written also IF NOT specified on QUESTION}}$$

$$= m_1 \ddot{y}_{G2} + m_3 \ddot{y}_{G3} + \underbrace{(m_1 + m_2 + m_3)g}_{\text{just constant term NOT affecting the SHAPE of diagram}}$$

We want to find

$\ddot{y}_{G2}$ and $\ddot{y}_{G3}$ explicitly.

looking @ V_{G2y} and V_{G3y} rowing (λ in) we can

$$\text{say } y_{G2} = \frac{L_2}{2} \theta \dots \text{MATLAB}$$

after computing V_0 we should define a **TIME HISTORY** for the harmonic output as function of time $\rightarrow$ so define a "TIME VECTOR"

if not provided by the text is free to choice

$$f \approx 1.6 \text{ Hz}$$

$$f \approx 4.3 \text{ Hz}$$

hence the excitation freq @ $\Omega_1 = 4 \text{ Hz}$
 $\Omega_2 = 3 \text{ Hz}$

it is enough
 $(0, 1) \text{ sec}$ time vector

totally 1 sec we
 get the period of both
 effect at least 2 period

get time

history result of $V_0(t)$ as harmonic output

(PART 2) effect of y_2 must be consistent with results of Lagrange equation

using super imposition so JUST HINGE constrain displacement $y(t)$

and $F(t) = 0$ NO SLIDER FORCE, only $y \neq 0$

only effect
 of displacem.
 forcing

$$(-\Omega^2 [M_{FC}] + i\Omega [C_{FC}] + [K_{FC}]) (\underline{x}_c)$$

forcing due to
 the $y(t)$ displacem.

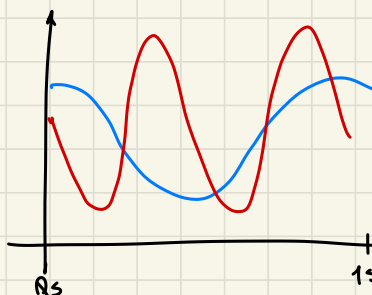
complex number
 i.e., $\underline{x}$ of horizontal
 movement

"
 $y(t)$ magnitude
 and phase!

than we

complat that

Time history respect the time vector



effect of f varying with 1 Hz
 impact we have 2 period of excitation

effect of $y(t)$ at 2 Hz so
 we see 2 period in 1s

we
 don't
 = STATIC

effect that

can be added later



when sum up the output
 you see that we get a

NO complex harmonic signal summing of two

$+ (m_1 + m_2 + m_3) * g$ Just shift vertically result

